

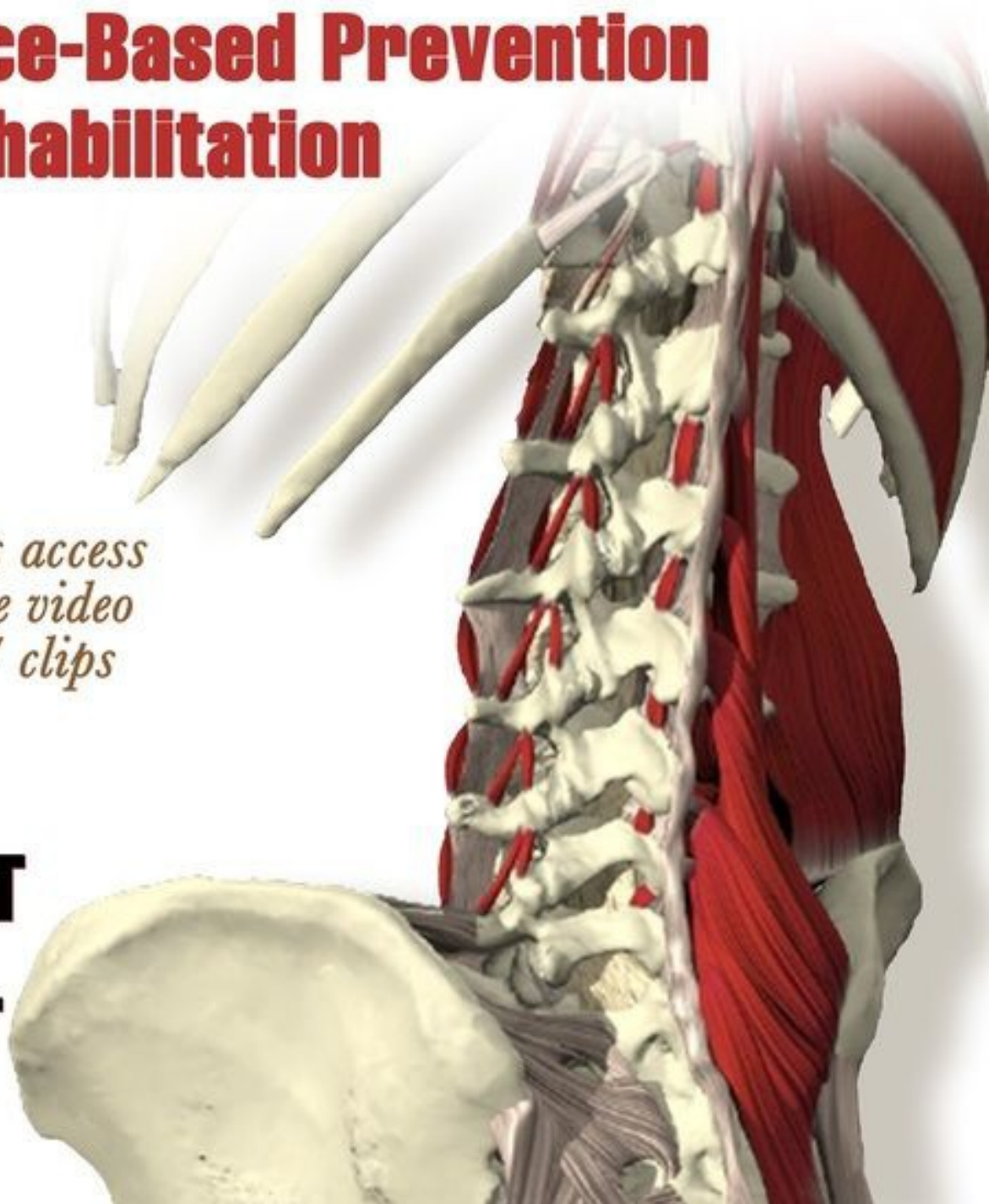
Third Edition

LOW BACK DISORDERS

**Evidence-Based Prevention
and Rehabilitation**

*Includes access
to online video
with 17 clips*

**STUART
McGILL**



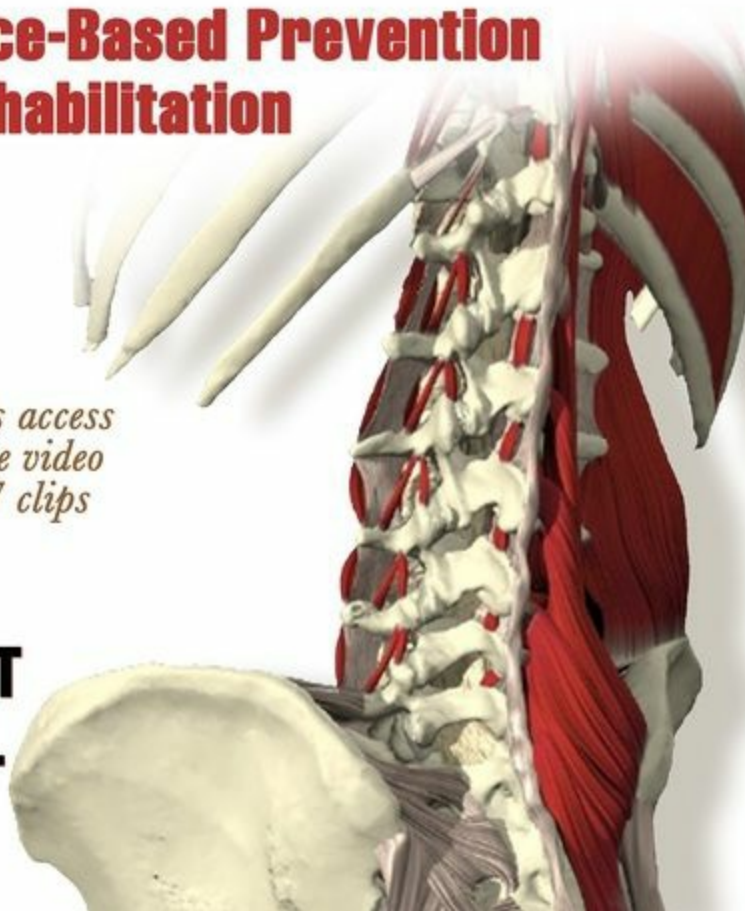
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Low Back Disorders

Evidence-Based Prevention and Rehabilitation

Third Edition

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Human Kinetics

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Check Out the Web Resource!

You will notice a reference throughout this version of Low Back Disorders, Third Edition, to a web resource. This resource is available to supplement your e-book. The web resource includes access to online video with 17 clips. We are certain you will enjoy this unique online learning experience.

Follow these steps to purchase access to the web resource:

1. Visit www.tinyurl.com/LowBackDisordersWR3E.
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3. After you have successfully completed your purchase, visit the book's website at www.HumanKinetics.com/LowBackDisorders.
4. Click the third edition link next to the corresponding third edition book cover.
5. Click the Sign In link on the left or top of the page and enter the e-mail address and password that you used during the purchase process. Once you sign in, your online product will appear in the Ancillary Items box. Click on the title of the web resource to access it.
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Click the Need Help? button on the book's website if you need assistance along the way.

I dedicate this book to my teachers. To my parents, John and Elizabeth, who gave what they could. To the few high school teachers who were able to see past my impatience and occasionally misguided enthusiasm—in particular, Ralph Colucci, who spent countless hours coaching on the track and football fields and teaching life’s lessons. He practiced the highest example of daily dedication and the drive to continue after others had given up. My university professors in the tough early years, particularly Dr. Klambauer, who was able to transform mathematics from abstract magic into something very mechanical that I could suddenly feel in my hands. This was a turning point. Professor David Winter, who taught me, among many other things, how to read a scientific paper. Professor Robert Norman, my PhD supervisor and mentor, who is the embodiment of integrity and the master of viewing any issue from the highest of intellectual vantage points. The many great academic personalities in the spine world with whom I have had discussions and debates, and whose writings I have studied, have enhanced my education and perspective—Harry Farfan, Bill Kirkaldy-Willis, Don Chaffin, Bill Marras, Nik Bogduk, Manohar Panjabi, Lance Twomey, and Mike Adams. And the academic clinicians who have stimulated me with their written and spoken thoughts—Vlad Janda, Shirley Sahrmann, Dick Erhart, Paul Hodges, Andry Vleeming, Craig Liebenson, Peter O’Sullivan, Gray Cook, Joachim Wilke, Robin McKenzie, and Clayton Skaggs. My many research colleagues from around the world, and visiting scholars to our lab, too numerous to mention here, who have taught me the many perspectives needed to temper the arrogance that comes so naturally when one is the only person privy to new research results. And to my wonderful graduate students—Jacek Cholewicki, Vanessa Yingling, Jack Callaghan, Sylvain Grenier, Lina Santaguida, Crisanto Sutarno, John Peach, Craig Axler, Lisa Brereton, Greg Lehman, Jennifer Gunning, Richard Preuss, Joan Scannell, David Bereznick, Kim Ross, Kelly Walker, Natasa Kavcic, Simon Wang, John Gray, Steve Brown, Janice Moreside, Sam Howarth, Leigh Marshall, Justin Yates, Stephanie Freeman, Rupesh Patel, Dave Frost, Dianne Ikeda, Christian Balkovec, Natalie Sidorkewicz, Ben Lee, Jordan Cannon, and Drs. Ed Cambridge, Doug Richards, and Claudio Tampier. Our lab technicians—Amy Karpowicz, and Chad Fenwick. The old dog has learned a few of your new tricks. And finally, to the many performance gurus with whom I have had many discussions and training sessions to try experimental approaches—these have assisted me in knowing what important aspects to test. Although it

is difficult to pick out a few, I will mention, in particular, the late Mel Siff, Juan Carlos Santana, Al Vermeil, Jerzy Gregorek, Pavel Tsatsouline, Mark McCoy, Art Horne, Dan John, Jon Chaimberg, and Bill Kazmaier—all immensely clever and generous men. For this third edition I also wish to thank many of you who have e-mailed me with your kind and encouraging words. All of you have contributed to my forging. Any remaining personal failings I can only attribute to the unfortunate combination of running Canadian software on Irish hardware.

Finally, to my wife and children, Kathryn, John, and Sarah, who have taught me to see the joy of the moment and enjoy the security of unconditional love. Quoting John Anderson, "I'm just a chunk of coal, but I'm gonna be a diamond some day!"

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Preface

Over the years since the first and second editions hit the bookstores, I have appreciated the testimonials and feedback from patients, clinicians, and educators who have used the book. This has motivated my research team to refocus on expanding our knowledge and best practice in several areas, which we were able to synthesize into an improved edition. Although updates are to be found throughout, major additions were made to several chapters, as noted here.

Ensuring a stable spine was the focus of the first edition. The second edition also addressed dealing with patients who have the opposite difficulty; that is, they have too much stiffness and crush their spines with chronic contraction, and are often locked in problematic postures. Yet others have regional stiffness, and most of the motion occurs at a single segment. We call this the spinal hinge. More often than not, these hinges are the site of pain. New studies have shown that these also predict where future back troubles will occur. Buttressing them with muscular straps or girdles is the key—and this is explained in this third edition.

Patient assessment has been extensively enhanced with more information about provocative testing. This will help you determine the cause of back troubles and, in turn, the best ways to eliminate the causes so that you can make better therapeutic exercise decisions. Our intention was to create a more thorough resource and reference for clinicians and savvy individuals. I have written a book for the lay public with back pain (*Back Mechanic: The Step-by-Step McGill Method to Fix Back Pain*) that is much more accessible to those unfamiliar with medical jargon. However, because back pain is often not simple to resolve, this resource remains essential.

Another emphasis of the first two editions was finding versions of stabilization exercises that could be conducted pain free. As new findings have emerged in our laboratory, we have been able to incorporate more information about tweaking the exercises to establish pain-free rehabilitation and training. Expansion of the pain-free motion repertoire is another critical component.

Too many patients believe that performing some exercises will take their pain away, and they simply add them to their daily routine. This is a mistake.

The first, and most critical, stage is to remove the cause of pain. This may be a poor choice of movement pattern that overstresses the painful tissue. More emphasis on choosing less stressful ways to move, together with reducing loading in the early stage of rehabilitation is provided in this latest edition. Also, the assessment section has been expanded.

The first edition was printed over a decade ago. Since that time several studies have tested the approaches we described and found them to be successful in reducing and eliminating pain in both individuals and populations such as the military. Some of this supporting evidence has been documented in this edition.

New to this edition is an accompanying web resource available at www.HumanKinetics.com/LowBackDisorders. The web resource includes video clips that showcase 17 exercises and tests explained throughout the book. These videos will help you apply the techniques discussed in this book to your own clients and patients. The web resource also provides blank copies of the handouts appearing at the back of this book. These handouts are presented as fillable PDFs that you may download and customize with instructions unique to patients' needs.

Also new to this edition is an image bank of the art, content photos, and tables used in the book. Instructors may access the image bank at www.HumanKinetics.com/LowBackDisorders and use the files to create customized presentations.

Finally, graduating into performance exercise is outlined in this edition. Although detailed data and guidance are contained in my other book, *Ultimate Back Fitness and Performance* (www.backfitpro.com), this text explains the critical stages of the performance pyramid:

1. 1. Design appropriate corrective exercise to address the causes of the disorder.
2. 2. Build joint and whole-body stability (and, by default, mobility).
3. 3. Enhance endurance.
4. 4. Train true strength.
5. 5. Transition to ultimate performance with superstiffness techniques, speed, power, and agility.

The back is not like any other joint or area of the body. It follows different rules—learn the rules and you will rule the back.

Accessing and Using the Web Resource

The web resource for Low Back Disorders, Third Edition, features 20 handouts in PDF format that clinicians may customize with detailed exercise instructions specific to individual clients. All of these handouts are accompanied by at least one photo showing a key movement of the exercise. The handouts can be printed out and distributed to clients for their use. In addition, the web resource features high-definition video of 17 of the tests and exercises described in the book.

To access the web resource, visit www.HumanKinetics.com/LowBackDisorders. If you purchased a new print book, follow the directions included on the orange-framed page at the front of your book. That page includes access steps and the unique key code that you'll need the first time you visit the Low Back Disorders, Third Edition, website. If you purchased an e-book from HumanKinetics.com, follow the access instructions that were e-mailed to you following your purchase.

To download and print out the handouts, select Web Resource in the ancillary items box in the upper left corner of the screen. You will now be taken to a page with two buttons: One for the handouts and one for the videos. Click the button for the handouts. At the bottom of the screen, select the link to open a handout.



To view the video clips, select Videos button on the Web Resource page. Select the link for the chapter you want. Once you select a chapter link, you'll see a video player. The video numbers along the right side of the player correspond with video number cross-references in the book, and the title under the player corresponds with the exercise title in the book. Scroll through the list of clips until you find the video you want to watch. Select that clip and the full video will play.



Here is a listing of the exercise handouts in the web resource, including an asterisk to denote the exercises that are demonstrated in video:

- Bird dog, remedial*
- Bird dog, beginner*
- Bird dog, intermediate*
- Bird dog, advanced*
- Bird dog, highest level*
- Cat–camel
- Curl-up, beginner*
- Curl-up, intermediate*
- Curl-up, advanced*
- Curl-up, highest level*
- Educating all back extensor muscle motor units
- Flossing, beginner*
- Isometric exercises for the neck
- Potty squat
- Remedial side bridge for deconditioned, chronic patient
- Remedial side bridge for painful shoulders 1
- Remedial side bridge for painful shoulders 2
- Side bridge, beginner
- Side bridge, intermediate

- Side bridge, intermediate variation
- Side bridge, advanced

Acknowledgments

No one will know the effort involved in writing a book until they have done so themselves. Many people have been part of the team. I specifically want to thank and acknowledge my editors at Human Kinetics. Fifteen years ago Loarn Robertson suggested that I write a book, which resulted in the first edition. Elaine Mustain masterfully grasped mechanical and medical concepts and helped guide my writing far beyond what was possible by my English teachers during the rewriting stage of the first and second editions. For this third edition, Melissa Zavala took over the editing duties that improved the extensive additions, while Neil Bernstein made me look good in the photos. We have all become good friends.

Why And How You Should Read This Book

Given the myriad stretch and strengthening programs available for the professional or layperson with a bad back, why write another book offering guidelines for better prevention and rehabilitation of low back troubles? The reason is that very few of the available books and manuals are actually based on scientific studies that investigate the mechanics of back function and the faulty mechanics that lead to pain. This has led to some programs recommending routines that replicate the mechanical causes of the damage! Such routines may actually increase the risk of developing back troubles or exacerbate existing ones. Other, more specialized books promote spine stabilization exercises without quantifying stability, thus basing the recommended exercises on a good guess—albeit a guess based on years of clinical observation. Scientific foundations to justify the recommendations have been thin at best. This book is not about perpetuating clinical myths—it is about challenging them and proposing valid and scientifically justifiable alternatives.

How the Book Is Organized

The intention of this book is to provide the best available scientific evidence to optimize injury prevention and rehabilitation efforts. It is designed to assist those who work to prevent low back troubles and those who are charged with the responsibility of rehabilitating them. Part I begins with a scan of the landscape related to the low back question. Here the main issues are introduced together with a brief introduction to the legislative and institutional constraints—some of which assist efforts to enhance back health, whereas others act as direct barriers. This is followed with evidence as to how the low back structure and associated tissues work during conditions of good health and how they become injured. These chapters describe the following:

- Injury mechanisms involving individual tissues
- Injury mechanisms involving the full lumbar mechanism
- Myths and realities of lumbar stability and mobility

Specific information—such as tissue loads, muscle activation levels, and measures of stability—is provided so that you can meaningfully compare approaches when deciding how to prevent or rehabilitate low back injury. Thus, part I provides the scientific foundation for the rest of the book.

Identifying and implementing the best programs result from considering the evidence. Thus, part II supports evidence-based practice by applying the scientific foundation in part I to the prevention of low back injury. Part III applies this same information to the rehabilitation of low back injury. No rehabilitation exercise program will be successful without removing the cause of the tissue sensitivity and pain.

Focus and Features of the Text

This text will be of value to those involved in designing and delivering back trouble prevention and rehabilitation programs—physical therapists, physiatrists, chiropractors, kinesiologists, ergonomists, exercise therapists, fitness experts, athletic trainers, coaches, and orthopedic specialists.

Low Back Disorders features a number of elements besides the basic text to ensure that all segments of the very diverse readership that deals with low back issues will be able to absorb and use the material. Special elements in some chapters highlight issues of methodological concern or simply provide further elucidation of some concepts or terms that are not common across disciplines. Terms that may be unfamiliar to some readers are defined in a glossary. In addition, a wide variety of occupational and athletic examples throughout the book can be generalized to all activities of daily living.

A major theme of this book is the benefits of blending rehabilitation efforts with ergonomics and prevention. Traditionally, ergonomists and industrial engineers have been responsible for prevention efforts mainly in the workplace, whereas the medical community has been responsible for rehabilitation. This natural division has caused an unfortunate compromise. Rehabilitation efforts will not be effective without preventing the cause of the tissue irritants, and prevention efforts will be retarded without an understanding of the variables obtained from medical investigations into the causes of tissue damage. No clinician will be effective if the cause of the patient's troubles is not removed. The expertise required for optimal prevention and rehabilitation can be found in the fields of ergonomics and rehabilitation medicine, which are naturally symbiotic. Many in the sport and business world have debated whether it is better to develop a stronger offense or defense. The obvious answer is that it is best to have both. I hope that this book will foster a better process resulting in more effective injury prevention and successful rehabilitation.

Some Encouraging Words

Some readers may be tempted to apply the material in parts II and III without digesting the foundation material contained in part I. This is a mistake. People who have attended my courses and were previously familiar with our published scientific papers often state that they were unable to fully synthesize the story and integrate the critical information. Once aware of the full story contained in the course and told in this book, they were much more effective at preventing low back disorders and prescribing exercises that help patients become better rather than worse.

A Note to Lay Readers

Although this book was written for clinicians, many laypeople have read it. As a result, they have written letters expressing their appreciation for their better understanding of the science and their own back troubles, and many have reported substantial improvement. However, we all recognize that having expertise enhances the reading experience of this book (I recently wrote *Back Mechanic: The Step-by-Step McGill Method to Fix Back Pain* for the lay public). This leads me to add that although successful self-treatment is gratifying, it removes some of the safeguards in the rehabilitation process. If you have a troubled back and are not a clinician, I remind you always to consult a physician at the outset to rule out any condition such as a tumor or a contraindication for exercise.